

Budget Calorie Meal Planner

Our project is a Budget Calorie Meal Planner written in Java. The purpose of the program is to help a user plan meals while keeping track of both their daily food budget and calorie intake.

When the program starts, the user enters their daily meal budget. The program checks the input to make sure the budget is a valid number greater than zero. If the user enters an invalid value, the program displays an error message and asks them to enter the budget again.

Next, the user selects either female or male so the program can determine the recommended daily calorie amount. The program uses 2,000 calories for women and 2,500 calories for men.

The user can then choose between breakfast, lunch, dinner, or finishing the day. Each meal has a set cost and calorie amount. Breakfast costs \$5 and contains 400 calories, lunch costs \$10 and contains 600 calories, and dinner costs \$15 and contains 900 calories.

Each time the user chooses a meal, the program checks whether there is enough money remaining in the budget. If the user can afford the meal, its cost is subtracted from the remaining budget, and its calories are added to the total calories consumed. The program continues allowing meal selections until the user chooses to finish or no longer has enough money for another meal.

Our program uses JOptionPane to make the program easier for the user to interact with through dialog boxes instead of requiring most information to be entered through the console. We also included input validation, loops, conditional statements, and a switch statement to control the different meal choices.

The main design goal was to create a simple program that combines budgeting with calorie tracking. Initial testing included entering different budgets, selecting different meal combinations, and entering invalid budget values to make sure the program responds correctly without crashing.

Overall, the program gives users a simple way to see how their meal choices affect both their budget and daily calorie intake.